Online Vehicle Rental System	Version: 1.1
System Specification	Date: 20/03/17

I. VEHICLE RENTAL SYSTEM SPECIFICATION

The Vehicle Rental store needs to build an online vehicle rental system with the following core functionalities:

- ***Browse and Search Vehicles:***
  - Customers can view vehicles by **vehicle type** (motorbikes, 4-seater cars, 7-seater cars, electric vehicles, etc.). The vehicle list can be sorted (ascending or descending) by criteria such as rental price, discount, and rental count.
  - Customers can search for vehicles by entering keywords. The system will return a list of vehicles whose name or model contains the keyword. The vehicle list can be sorted (ascending or descending) by rental price, discount, and rental count.
- ***Select and Book Vehicles:***
  - Each **vehicle** has a vehicle name, unique vehicle code, license plate, illustrative images, summary and detailed descriptions (make, seat capacity, fuel type, transmission), daily rental price, availability status (available or rented), and view count. Each vehicle belongs to a specific **vehicle category**.
  - When browsing or searching for vehicles, for each item in the list, customers can *add the selected vehicle to the current rental cart*, or *view detailed information about the vehicle*. While viewing details, customers can also *add the vehicle to the rental cart*.
  - During the selection process, customers can *view the list of selected vehicles in the cart*, *update the rental cart* (remove vehicles, update rental duration and vehicle quantities).
  - Customers select the **booking function** after finishing their vehicle selections to officially place a rental reservation.
  - **Vehicle Booking Process:**
    - The system will prompt the customer to **log in** if not logged in, or **register a new account** if they do not have an account yet.
    - Customers enter the **recipient/driver information** (full name, address, phone number, driver's license number). Note that the driver/recipient can be different from the buyer (e.g., booking for a colleague or relative).
    - If the rental order value is **2,000,000 VND** or more, delivery is free. Otherwise, the system calculates a **delivery fee** depending on the **delivery area** (based on recipient address down to ward level) and selected delivery method. The delivery fee (if any) is added to the invoice total.
    - Customers can choose cash payment upon vehicle handover or online payment. *If online payment is chosen, the system automatically connects to the Online Payment Service Gateway*.

SPECIFICATION (DRAFT)

- Upon successful booking, the system will **record the rental order**, including vehicle items, *quantity* and corresponding *unit price*, renter and recipient info, delivery method, total order value, timestamp, and payment method.
  - If an email is provided, the system will **send a booking confirmation email** with full order details.

- ***Customer Registration & Profile Updates:***
  + Customers can register an account in the system.
  + Customer information includes: full name, date of birth, gender, address, phone number, password, email address (used as login username, order notifications, and password recovery).
  + Customers can update all their profile information except their username (email).
  + Forgotten Password case: Customers use the password recovery function; the system generates a new password and sends it to their email.
- ***View Placed Rental Orders:***
  + Customers can review placed rental orders and their status (New, Confirmed, Delivering, In-Rental, Completed, Cancelled).
  + Customers can cancel an order if its status is "New".
- ***Staff Operations:***
  + View rental orders and update processing status.
  + Perform vehicle handover and vehicle return inspection procedures.
- ***System Administration:*** Administrators have the following functions:
  + Manage vehicles and vehicle categories
  + Manage users (Customers & Staff)
  + Manage rental orders

II. GLOSSARY

1. **Discount:** Calculated as a percentage (%) of the current rental price.
2. **Rental Cart:** List of vehicles selected by the customer prior to order confirmation/deposit.
3. **Vehicle Category:** Classification of vehicles (Motorbikes, 4-seater cars, 7-seater cars, Electric vehicles, etc.).

SPECIFICATION (DRAFT)